Business Problem Statement

Customer Shopping Behavior Analysis | Python • SQL • Power BI

The Business Problem

A retail company sells clothing, accessories, footwear and outerwear to customers across the United States. Management wants to understand **why** spending, loyalty and satisfaction differ between customer groups. Decisions on discounts, subscriptions, shipping options and seasonal stock need to be backed by evidence of which factors actually grow revenue and bring customers back.

Management is particularly interested in understanding how discounts, review ratings, seasons, shipping choices, subscription status and payment preferences influence what customers buy, how much they spend and whether they return.

Overarching Business Question

> *"How can the company use its customer shopping data to understand who its most valuable customers are, what drives their purchasing decisions, and how to improve engagement, loyalty and revenue through smarter marketing and product strategies?"*

About the Dataset

The analysis uses a customer shopping dataset of **3,900 records (one row per unique Customer ID) and 18 attributes**, covering customers aged 18 to 70 from all 50 U.S. states:

Customer profile	Customer ID, age, gender, location (50 states), subscription status, previous purchases (1 to 50)
Product details	Item purchased (25 items across 4 categories: Clothing, Accessories, Footwear, Outerwear), size, colour, season
Transaction behaviour	Purchase amount (USD), discount applied, promo code used, shipping type (6 options), payment method (6 options), frequency of purchases
Customer feedback	Review rating (37 missing values require treatment)

Key Questions to Answer

Revenue & Customers	Who generates the most revenue by gender, age group and location? Do subscribers spend more than non-subscribers?
Loyalty & Retention	How do New, Returning and Loyal customers compare? Are repeat buyers more likely to subscribe? Does purchase frequency affect total spend?
Discounts & Pricing	Which products rely most on discounts? Do customers who use discounts still spend above the average purchase amount?

Products & Seasons	Which products and categories sell best, and which earn the highest ratings? Which season drives the most revenue? What sizes and colours are most popular?
Operations & Payments	How does spend differ across shipping types, including Standard and Express? Which payment methods generate the most revenue?
Satisfaction	How do review ratings differ across subscription status, categories and products?

Deliverables

1. Data Preparation & Modeling (Python)

Load the raw dataset with pandas, inspect its structure, and handle missing review ratings (median imputation by product category). Standardise column names to snake_case, engineer new features (*age_group*, *purchase_frequency_days*), remove redundant fields such as *promo_code_used*, and explore the data visually before loading it into a MySQL database.

2. Data Analysis (SQL)

Organise the cleaned data into a structured *customer* table in MySQL and write queries to extract insights on customer segments, loyalty, discount behaviour, product performance and purchase drivers, using aggregations, subqueries, CTEs and window functions.

3. Visualization & Insights (Power BI)

Build an interactive dashboard that highlights key patterns and trends across customer segments, products, seasons and payment methods, enabling stakeholders to make data-driven decisions.

4. Report and Presentation

Write a clear project report summarising key findings and business recommendations, and prepare a presentation that visually communicates insights and actionable recommendations to stakeholders.

5. GitHub Repository

Publish all Python scripts and notebooks, SQL queries, and dashboard files in a well-structured repository with a README.